

Question	Marking Guidance	Mark	Comments
10.1	1. (Refers to) saltatory conduction OR (Nerve) impulses/depolarisation/ions pass to other neurones OR Depolarisation occurs along whole length (of axon); 2. (Nerve) impulses slowed/stopped; 3. (Refers to) <u>neuromuscular</u> junction OR (Refers to) <u>sarcolemma</u>;	3	1. Accept suitable description that refers to (transmission) from node to node (of Ranvier). 1 and 2. Accept action potentials for impulses. 1. Accept action potential for depolarisation. 1, 2 and 3. Reject first mark awarded if answer refers to messages/signals for impulses. Reject even if impulse/s also referred to.
10.2	1. Slower/fewer impulse(s) along sympathetic/parasympathetic (pathway/neurones); 2. (Impulses) from cardiac centre OR (Impulses) from medulla; 3. To SAN;	3	1. Accept action potentials for impulses. 1.Reject no impulses. 1, 2 and 3. Ignore 'information' but reject first mark awarded if answer refers to messages/signals for impulses. Reject even if impulse/s also referred to.
10.3	1. It/DNA is <u>complementary</u> to (m)RNA; 2. Binds to mRNA (for huntingtin); 3. Prevents <u>translation</u>;	3	1. Accept (transcription) results in <u>complementary</u> (m)RNA. Ignore miRNA/siRNA/transcriptional factors. 3. Ignore transcription.

10.4	1. Small sample size OR Only 46; 2. Only four-months OR short period (of trial); 3. Huntingtin/protein reduced OR Huntingtin/protein still produced OR Huntingtin/protein not removed; 4. Allele/gene/mutation/mRNA (for Huntington's) still present OR (May be) temporary OR Drug treatment repeated; 5. Brain already damaged OR Brain damage may continue;	2 max	3. Accept huntingtin for huntingtin. Ignore miRNA/siRNA/transcriptional factors.
10.5	1. (Drug/DNA) will directly/quickly reach <u>brain</u> OR (Cerebrospinal) fluid bathes the <u>brain</u>; 2. (Drug/DNA) not destroyed by acid OR (Drug/DNA) not digested (by enzymes);	2	2. Reject protein is digested. 2. Ignore location of enzymes. 2. Accept Drug/DNA denatured.
10.6	1. (Increased) methylation of DNA/gene/allele; 2. Inhibits/prevents transcription; OR	2 max	Mark in pairs but if no mark credited allow one mark for any reference to transcription or gene expression being affected.

	3. Decreased methylation of DNA/gene/allele; 4. Stimulates/allows transcription; OR 5. Decreased acetylation of <u>histone(s)</u>; 6. Inhibits transcription; OR 7. Increased acetylation of <u>histone(s)</u>; 8. Stimulates/allows transcription;		1. Reject acetylation of DNA. Accept gene expression for transcription but ignore gene switched on/off. Ignore methylation of histones. Accept DNA-histone complex as equivalent to <u>histone(s)</u>. .
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